9A Lesson 5

Genetic counselling

Year 9 Science · 9A Genetics and evolution · Student Book pages 14 to 15

Name _____ Class _____ Date _____

Today I can:

☐ I can describe what a genetic test looks at and why babies and families may be tested.

☐ I can use probability and my knowledge of genes and chromosomes to explain a genetic test result simply, as a genetic counsellor would.

Key words

genetic test a test that looks at a person's DNA sequence or chromosomes

genetic counsellor a person trained to explain genetic test results and the chances of passing on a condition

genetic disorder a condition caused by a faulty gene or an unusual set of chromosomes

DNA sequence the order of the four letters (A, T, C, G) along a DNA molecule

faulty gene a gene whose sequence has changed so that it no longer works properly

haemoglobin the protein in red blood cells that carries oxygen

sickle cell disease a genetic disorder in which faulty haemoglobin makes red blood cells sickle-shaped

cystic fibrosis a genetic disorder in which thick mucus builds up in the lungs and gut

Down's syndrome a condition caused by an extra copy of chromosome 21

probability the chance of something happening, e.g. 25% or 1/4

chromosome a structure in the nucleus made of one long DNA molecule

● Core Everyone does this section.

1. People P, Q, R and S had the same gene analysed. Circle the group of letters that is different, and state which person has a gene that is not the same as the others.

P CTT CTC GGA TCG CGG GAA ATA TTC

Q CTT CTC GGA TCG CGG GAA ATA TTC

R CTT CTC GGA TCG CCG GAA ATA TTC

S CTT CTC GGA TCG CGG GAA ATA TTC

The person with the different gene is _____ .

2. Match each term to its meaning. Draw a line.

genetic test	a condition caused by a faulty gene or unusual chromosomes
genetic counsellor	the chance of something happening
genetic disorder	a test that looks at a person's DNA or chromosomes
probability	a person who explains genetic test results and chances to families

3. The different version of the gene in question 1 causes a disorder in which waste substances cannot be removed from the body. Explain why finding out that a baby has this version of the gene is better than waiting until the baby is a child.

4. A test report reads: 'Both parents have a faulty version of the CFTR gene. There is a 25% chance of them having a child with cystic fibrosis.'

a) If the parents have four children, how many are predicted to have cystic fibrosis?

b) Explain why the real number could be different from your prediction.

●● Stretch When Core is finished.

S1. A chromosome test showed that a baby's body cells each contain 47 chromosomes: three copies of chromosome 21 instead of two. This causes Down's syndrome.

a) State the number of chromosomes in a normal human body cell and in a normal gamete.

b) Suggest how a zygote could end up with three copies of chromosome 21. (Hint: think about what each gamete carries.)

S2. A test report reads: 'The patient has one copy of the E4 variant of the APOE gene. The probability of developing Alzheimer's disease before the age of 85 is 20% (compared with 13% for the general population).' Write what you would say to explain this simply to the patient. Use 'out of 100'.

●●● Challenge Explain in full sentences.

C1. A person's chromosome test shows an unusual arrangement of two chromosomes. The person has no health problems. A woman with chromosomes like this has a 10% chance of having a baby with Down's syndrome; a man with the same chromosomes has no increased risk. You are the genetic counsellor. The person tells you they want to start a family. Write what you would say. Include the probability, what it means, and what the options are.

C2. Suggest why genetic tests on babies are becoming more common, and give one argument against testing every baby for every known disorder.

My thinking (any language)

If a test could tell you your chance of getting a disease with no cure, would you want to know? Explain your thinking.

Word bank genetic test · genetic counsellor · genetic disorder · DNA sequence · faulty gene · haemoglobin · probability · chromosome · cystic fibrosis · Down's syndrome

✂ ---------- cut here

Exit ticket 9A Lesson 5: Genetic counselling Name _____

1. What does a genetic test look at?

2. Both parents carry a faulty gene and each child has a 25% chance of the disorder. Write 25% as a fraction and as a decimal.

3. Give one thing a genetic counsellor does.

Today I can: Goal 1 ☐ Goal 2 ☐